WORKED SOLUTION

STEP I 1988, Question 7

Pure • Bounding Quadratic Derivatives

STEP I 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q7

This question has a full guided treatment in *Beyond STEP*, Section 3.6, *A Bound on Slopes*: https://beyondstep.keithmcnulty.org/ch03/markov-bound.html

The key idea

A quadratic is pinned down by its values at three points, so f' can be rebuilt from $\mathrm{f}(1)$, $\mathrm{f}(-1)$ and $\mathrm{f}(0)$. Once f' is written that way, the hypothesis $|\mathrm{f}| \le 1$ controls each of the three coefficients, and the triangle inequality does the rest. For the final part, we look at which values of f make every inequality in the proof an equality.

Solution

The identity for $\mathrm{f}'(x)$

We have $\mathrm{f}(1) = a+b+c$, $\mathrm{f}(-1) = a-b+c$ and $\mathrm{f}(0) = c$. Collecting the right-hand side by powers of x:

$$\begin{aligned}&\mathrm{f}(1)\left(x+\tfrac{1}{2}\right) + \mathrm{f}(-1)\left(x-\tfrac{1}{2}\right) - 2\mathrm{f}(0)x \\ &\quad = \big[(a+b+c)+(a-b+c)-2c\big]x + \tfrac{1}{2}\big[(a+b+c)-(a-b+c)\big] \\ &\quad = 2ax + b.\end{aligned}$$

Since $\mathrm{f}'(x) = 2ax+b$, this is the required identity.

The bound $|\mathrm{f}'(x)| \le 4$

Suppose $|\mathrm{f}(x)| \le 1$ whenever $|x| \le 1$. In particular $|\mathrm{f}(1)|$, $|\mathrm{f}(-1)|$ and $|\mathrm{f}(0)|$ are all at most 1. Take any x with $|x| \le 1$. By the identity and the triangle inequality,

$$\begin{aligned}|\mathrm{f}'(x)| &\le |\mathrm{f}(1)|\,\big|x+\tfrac{1}{2}\big| + |\mathrm{f}(-1)|\,\big|x-\tfrac{1}{2}\big| + 2|\mathrm{f}(0)|\,|x| \\ &\le \big|x+\tfrac{1}{2}\big| + \big|x-\tfrac{1}{2}\big| + 2|x|.\end{aligned}$$

Ask yourself How big can $\left|x+\frac{1}{2}\right|+\left|x-\frac{1}{2}\right|$ be when $|x|\le 1$?

This is the sum of the distances from x to $\frac{1}{2}$ and to $-\frac{1}{2}$. If $|x|\le\frac{1}{2}$, then x lies between the two points and the sum is exactly 1. If $|x|>\frac{1}{2}$, both points lie on the same side of x, and the sum is $2|x|$. In either case it is at most 2 for $|x|\le 1$. Also $2|x|\le 2$, so

$$|f'(x)|\le 2+2=4 \qquad \text{for } |x|\le 1,$$

as required.

(Another way to see this: f' is linear, so on $[-1,1]$ its modulus is largest at an endpoint, and the bound at $x=\pm1$ reads $|f'(\pm1)|\le\frac{3}{2}+\frac{1}{2}+2=4$.)

A case of equality

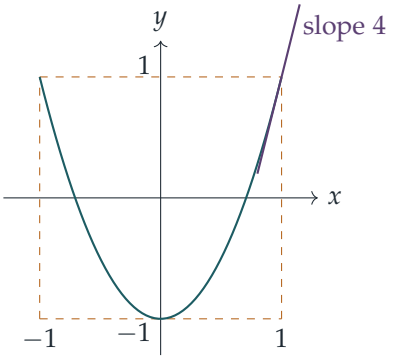
For equality at $x=1$ we need every step above to be tight there. At $x=1$ the coefficients $x+\frac{1}{2}$, $x-\frac{1}{2}$ and $-2x$ are $\frac{3}{2}$, $\frac{1}{2}$ and -2. To get $f'(1)=+4$ we therefore want

$$f(1)=1, \qquad f(-1)=1, \qquad f(0)=-1.$$

These give $c=-1$, $a+b+c=1$ and $a-b+c=1$, so $b=0$ and $a=2$:

$$a=2, \quad b=0, \quad c=-1, \qquad f(x)=2x^2-1.$$

Now check the condition. For $|x|\le 1$ we have $0\le x^2\le 1$, so $-1\le 2x^2-1\le 1$, that is, $|f(x)|\le 1$. And $f'(x)=4x$, so $f'(1)=4$ with $x=1$ in the interval. This is the example we need.



The curve $y=2x^2-1$ (teal) stays inside the square $|x|\le 1$, $|y|\le 1$, swinging from top to bottom and back, and so it is as steep as it can be at the ends.

(Any of $\pm(2x^2-1)$ gives $|f'(\pm1)|=4$. The choice $f(x)=2x^2-1$ gives $f'(1)=+4$, and so does $-(2x^2-1)$ at $x=-1$.)

Checks

- Try $f(x)=x^2+x+1$: $f(1)=3$, $f(-1)=1$, $f(0)=1$. At $x=2$ the identity gives $3\cdot\frac{5}{2}+1\cdot\frac{3}{2}-2\cdot 2=5=f'(2)$.

- A random search over 20,000 quadratics scaled so that $\max_{|x|\le 1} |f(x)| = 1$ never found $\max_{|x|\le 1} |f'(x)| = 2|a| + |b|$ above 4 (the largest was about 3.91), and $2x^2 - 1$ reaches 4 exactly.

Common slips

- Bounding $\left|x + \frac{1}{2}\right|$ and $\left|x - \frac{1}{2}\right|$ separately by $\frac{3}{2}$ each. That only gives $|f'(x)| \le 5$. The two distances must be bounded together.
- Proposing an example without checking $|f(x)| \le 1$ on the whole interval, not just at 0 and ± 1.

Beyond STEP

This is the degree-two case of Markov's inequality: if a polynomial p of degree n has $|p| \le 1$ on $[-1, 1]$, then $|p'| \le n^2$ there. Equality comes from the Chebyshev polynomial $T_n(\cos\theta) = \cos n\theta$, and $2x^2 - 1 = T_2(x)$. Writing a polynomial through its values at chosen points is Lagrange interpolation, a basic tool in numerical analysis and approximation theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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